

SONA CATHERINE

sonacatherine06@gmail.com | <https://github.com/Sonacatherine06> | www.linkedin.com/in/sona-catherine/ | 6282654154 | Kerala

Final year Electronics and Communication Engineering student at Christ College of Engineering, Irinjalakuda. Exploring embedded systems, IoT, and electronics through academic projects and practical learning experiences. Gaining hands-on experience through internships, workshops, and technical projects while developing problem-solving and teamwork skills.

INDUSTRY INTERNSHIPS

Indian Institute of Information Technology (IIIT) Kottayam

July 2026

- Completed training in **Internet of Things (IoT)** and **Robotics**.
- Developed practical knowledge of embedded systems, sensors, and actuators.
- Worked on basic automation and IoT-based projects.
- Improved skills in hardware integration and problem-solving.

BSNL Regional Telecom Training Centre

June 2026

- Successfully completed a two-week internship at **BSNL Regional Telecom Training Centre**.
- Gained practical knowledge of **telecommunication systems** and **network technologies**.
- Learned the fundamentals of **optical fiber communication** and its applications.
- Developed an understanding of basic telecom infrastructure and communication networks.

Evolve Robotics

June 2025

- Completed a two-week internship at **Evolve Robotics** in Embedded Systems.
- Worked with the **PIC16F877A** microcontroller for hardware programming.
- Performed circuit simulation using **Proteus Design Suite**.
- Gained practical experience in **Embedded C** programming using **MPLAB IDE**.

PROJECTS

- **Alcohol Detection with Engine Lock System**

Developed an alcohol detection and engine lock system using an alcohol sensor and microcontroller to prevent vehicle ignition when alcohol levels exceeded a predefined threshold. Programmed the microcontroller in Embedded C to process sensor data and control the ignition locking mechanism, enhancing vehicle safety through automated intoxication detection.

- **Muscle Translation Device for Paralyzed Individuals**

A biomedical assistive which is processed by an STM32 microcontroller with firmware programmed in STM32CubeIDE system, converts muscle signals into useful actions. An EMG sensor is used to capture muscle signals.

- **Smart Reminder system for paralysed patients**

developed a smart reminder system using the ESP32 microcontroller for scheduling and alert notifications. Programmed the ESP32 using Embedded C in Arduino IDE to manage reminder events and user inputs. Integrated a buzzer and display module to provide real-time visual and audible alerts. Designed and implemented an LDR-based automatic lighting system that automatically controls lighting based on ambient light intensity using sensor-based switching

CERTIFICATIONS & ACHIEVEMENTS

- MATLAB Onramp Certification – MATLAB Academy (2025)
- NPTEL Certification: Design Thinking – A Primer (2025)
- PCB Design and Fabrication Workshop – Neo Green Labs (2024)
- H2OForge Underwater Robotics Workshop – IEEE Computer Society Kerala Chapter (2023)
- First Prize, Spacecraft Model Making Competition, National Space Day 2024 (ISRO)

EDUCATION

- **B.Tech in Electronics & Communication Engineering— Christ College of Engineering, Irinjalakuda** **2023-2027**
 - CGPA (up to S5): 6.73/10
 - **Higher Secondary— St. Mary’s HSS, Irinjalakuda** **2022-2023**
 - Biology Science
 - Percentage: 90.75%
-

TECHNICAL SKILLS

- C Programming
- Fundamental Assembly Language Knowledge
- Basic proficiency in Verilog HDL
- Embedded systems basics